Every one-loop component on all local operators from the defect flow: the strict Costello route, operator side

Step 2, strict route (R10), operator theory

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Abstract

For every generator x of $\mathfrak{psl}(4|4)$ – the dilatation Δ, the translations P_b, the rotations of V, the rotations of $W_{\mathbb{R}}^{\perp}$, the supercharges $\xi \in \Sigma$ of both chiralities, the special supercharges $\eta \in \Sigma^*$ and the special conformal fields K_b – the one-loop component $\hbar L_1(O)^{(1)}(x)$ of every local functional O, gauge-variant ones included, is derived as the $\varepsilon \to 0$ limit of the graphs of the defect flow of Lemma c8-transfer, read through the local representative of Definition c8-rep. The graphs are those with one special edge $K_\varepsilon - K_0$, one vertex $S^{(1)}(x)$ in every placement (on a line, at the special edge, cubic), the composite lines, the finite special vertices, the counterterm of O and the arity-one counterterm; the one-body part is the bracket with the arity-one finite counterterm $\hbar F_1^{(1)}(x)$ of the action-side solution of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$, taken as an input with its free directions. The identities that earlier work used as inputs are proved here as consequences of the construction: the decomposition of the dilatation component into four times the natural logarithmic map, the composite pair kernels and a one-body term; the vanishing variation for translations and rotations; the position decompositions along η and K_b; and $K = P^\dagger$ for the one-loop radial inner product, with its hypotheses. The two-body locality of the composite term is proved completely, including the cancellation of its nonlocal one-body parts on the letters $A|_V$, λ, c, and its form for gauge-variant operators. The families on which every component is to be evaluated, the one-loop algebra on operators that follows from the arity-two master equation, with its form as a Lie superalgebra one-cocycle on cohomology, and the specification of two methodologically independent engine routes with their gates (`engine_spec.json`) complete the note. No dimensional regularisation enters in any role; the only infrared datum is the transfer endpoint T.

Contents

1 Data, generators and the input from the action side **1**
1.1 The generators on the regulator . . . 2
1.2 The input from the action side . . . 4
1.3 Defect data of a pair . . . 5

2 The components from the defect flow **6**
2.1 The defect vertex of an operator . . . 6
2.2 The master formula . . . 6
2.3 The Ward identity of the flow along an even generator . . . 9
2.4 The second evaluation and the identities along the even generators . . . 10

3 Position decompositions and $K = P^\dagger$ **12**
3.1 Position decompositions . . . 12
3.2 The special conformal component at a finite endpoint . . . 14
3.3 The conjugation and $K = P^\dagger$ . . . 14

1 Data, generators and the input from the action side

All conventions are those of the manuscript, Chapters 1–11, and of the notes `step2/twoletter/theory/twoletter_theory.tex`, `step2/composite_qsk/composite/theory/composite_theory.tex`, `step2/composite_qsk/qsk/theory/qsk_theory.tex`. References written as Lemma c8-transfer, Def c8-rep, Thm c11-local are to labels of the manuscript; references with numbers are internal. The weight is $e^{S/\hbar}$, $S=\frac12\langle e,Qe\rangle+I$, on arity zero $s=Q+\{I,-\}$, and on cochains $s=d_{CE}+Q+\{I_{\mathfrak{psl}(4|4)},-\}$ (Thm c8-obstruction). Edges carry $P(\varepsilon,T)=\int_\varepsilon^T(Q^{GF}\otimes 1)K_t\,dt$ (Def c8-PT), represented by the operator $P\star=Q^{GF}g(H)$, $g(H)=\int_\varepsilon^T e^{-tH}dt$, with

$$Q\,P\star+P\star\,Q=E(H):=e^{-\varepsilon H}-e^{-TH}. \tag{1.1}$$

The special edge is $K_\varepsilon-K_0=\big((e^{-\varepsilon H}-1)\otimes 1\big)K_0$, the factor $e^{-\varepsilon|p|^2}-1$ times the components of K_0, field slot against antifield slot (Prop c4-defect1). The endpoint T is the endpoint of the transfer of [C07]; every component below is a local functional independent of T (Thm c8-L (ii)); it is evaluated at $T=\infty$ where the graph sums converge at generic momenta (Prop c8-Tsafe) and at a finite T along K_b (Def c8-K). No dimensional regularisation is used, as a scheme, as an intermediate step or as a check. The unit is

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}=-\frac{g^2N}{32\pi^2}\qquad(\mathfrak g=\mathfrak{su}(N)),$$

written "units" after a number. The field-counting functional is written

$$\mathcal N:=2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),$$

whose bracket multiplies the coordinate functionals of $c,A,\lambda,A^*,\lambda^*,c^*$ by $1,2,4,-2,-4,-1$ (Lemma c6-count); the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ is s of the unit times $\mathcal N$ (Thm c4-log).

Letters and pair data. Letters, slots, Weyl weights, degrees, pair weight, pair degree and Koszul signs are those of Def c10-letters and eq c10-decomp; the slots are c, $A|_V$, $A|_{W_{\mathbb R}^\perp}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$, $A^*|_V$, $A^*|_{W_{\mathbb R}^\perp}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, of Weyl weights $0,1,1,\frac32,\frac32,3,3,\frac52,\frac52,4$ and degrees $1,0,0,0,0,-1,-1,-1,-1,-2$. For a local functional M the arity-zero counterterm is

$$\partial_\delta I^{CT}[\varepsilon](M)=\sum_{m\ge1}\varepsilon^{-m}\mathrm{ct}_m(M)+\log\varepsilon\,\mathrm{lc}(M),$$

two-body by Thm c10-decomp, and the natural logarithmic map is the map of Thm c10-brst,

$$\mathrm{lc}^{\mathrm{nat}}(M):=\mathrm{lc}(M)-\{\mathrm{unit}\cdot\mathcal N,\,M\}, \tag{1.2}$$

which commutes with s. For a monomial $O=\int\omega\,X_1\cdots X_n$ and a two-body kernel κ we write

$$\sum_{a<b}\pm\kappa(X_a,X_b):=\sum_{1\le a<b\le n}(-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)}\int\omega\,X_1\cdots X_{a-1}\,\kappa(X_a,X_b)\,X_{a+1}\cdots\widehat{X_b}\cdots X_n, \tag{1.3}$$

the sum over the unordered pairs with the Koszul signs of eq c10-decomp.

1.1 The generators on the regulator

Definition 1.1 (Generators and their operators)**.** The generators are, with their ad_Δ eigenvalue λ_x and the degree of their Killing data in the position:

generator x	Killing data	λ_x	degree in the position
Δ	$v = 2E$	0	1
P_b, $b \in V$	$v = b$	-2	0
M_ω, $\omega \in \mathfrak{so}(V)$	$v(x) = \omega x$	0	1
$R \in \mathfrak{so}(W_{\mathbb{R}}^{\perp})$	rotation of the index of $\iota_w A$ and its spin lift on λ	0	0
$\xi \in \Sigma = S_+ \otimes W \oplus S_- \otimes W^*$	constant Killing spinor ξ	-1	0
$\eta \in \Sigma^*$	Killing spinor $\rho(x)\eta$	$+1$	1
K_b, $b \in V$	$v(x) = (x,x)b - 2(b,x)x$	$+2$	2

For an even generator $x.$ is the operator $v.$ of Def c7-vdot on all slots (for R the rotation of $W_{\mathbb{R}}^{\perp}$, Prop c7-cov); for an odd generator it is the odd operator that induces the bracket with the quadratic part of $S^{(1)}(x)$ (Lemma c11-lines). With these, $S^{(1)}(x) = \int \big((x.A, A^*) - (x.\lambda, \lambda^*) - (x.c, c^*)\big)$ for even x, and for odd x the quadratic part of $S^{(1)}(x)$ is induced by $x.$ while $S^{(1)}(x)$ has the cubic terms T_5 of Table c11-terms. Two-component: P_b is $b^m\partial_m$, K_b has $v^m = x^2 b^m - 2(b\cdot x)x^m$, M_ω has $v^m = \omega^m{}_n x^n$, the Killing spinor of $\xi + \rho(x)\eta$ is eq c8-killing, $\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i\,x_m \sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i\,x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}$, and R acts on A_{ij} and $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ through the matrix $R^i{}_j$ of $\mathfrak{su}(4)$, $A_{ij} \mapsto -R^k{}_i A_{kj} - R^k{}_j A_{ik}$, $\lambda_\alpha{}^i \mapsto R^i{}_j \lambda_\alpha{}^j$, $\bar\lambda_{\dot\alpha i} \mapsto -R^j{}_i \bar\lambda_{\dot\alpha j}$.

Definition 1.2 (Variation kernel)**.** For a generator x and $0 < \varepsilon < T \le \infty$ the variation kernel of the propagator is

$$\mathcal{V}_x(\varepsilon, T) := (x. \otimes 1 + 1 \otimes x.)\, P(\varepsilon, T).$$

Lemma 1.3 (Commutators and variation kernels)**.** *For every generator, $[x., Q] = 0$, $x.$ preserves the pairing $\langle\,,\rangle$, and $\mathcal{V}_x(\varepsilon, T) = \big([x., P\star] \otimes 1\big)K_0$, the graded commutator. Moreover:*

- *(i) Δ: $[v., Q^{GF}] = -4Q^{GF}$, $[v., H] = -4H$, $\mathcal{V}_\Delta(\varepsilon, T) = -4(\varepsilon\partial_\varepsilon + T\partial_T)P(\varepsilon, T)$.*
- *(ii) P_b, M_ω, R: $[x., Q^{GF}] = [x., H] = 0$ and $\mathcal{V}_x = 0$; the same holds for K_t and for the special edge.*
- *(iii) $\xi \in \Sigma$: $[\xi., H] = 0$, $C_\xi := [\xi., Q^{GF}] \neq 0$ is the even, constant-coefficient operator of order one of Lemma c11-lines (ii), and $\mathcal{V}_\xi = \big(C_\xi\, g(H) \otimes 1\big)K_0$, with components in $\lambda \otimes A|_V$, $c \otimes \lambda^*$ and their transposes.*
- *(iv) $\eta \in \Sigma^*$: $\eta. = \sum_\mu x^\mu\, \xi_\mu. + \eta^0. + Y_\eta$ with $\xi_\mu = \rho(e_\mu)\eta$, $[\eta., H] = 2\sum_\mu \xi_\mu.\,\partial_\mu$, and*

$$[\eta., P\star] = \sum_\mu x^\mu\, C_{\xi_\mu} g(H) + \sum_\mu [x^\mu, Q^{GF}]\, \xi_\mu.\, g(H) + [\eta^0. + Y_\eta, Q^{GF}]\, g(H) + 2\, Q^{GF} \sum_\mu \xi_\mu.\,\partial_\mu\, g'(H).$$

- *(v) K_b: $[v., Q^{GF}] = 4(b,x)Q^{GF} + C_K$, $[v., H] = 4(b,x)H - 4(w-1)(b,\partial) + 4b_\nu \Sigma^{\rho\nu}\partial_\rho$ on a slot of Weyl weight w (Lemma c11-lines (v)), and $\mathcal{V}_{K_b} = ([v., P\star] \otimes 1)K_0$ with $[v., P\star] = [v., Q^{GF}]g(H) + Q^{GF}[v., g(H)]$, $[v., g(H)] = \int_\varepsilon^T \int_0^t e^{-(t-t')H}[v., -H]e^{-t'H} dt' dt$.*

Proof. $[x., Q] = 0$ and the preservation of the pairing are Prop c7-cov, Prop c7-dil, Prop c7-susy and Lemma c11-lines (iv); they encode the parts of polynomial degree two of $\{S, S^{(1)}(x)\} = 0$ (Lemma c2-components). For an operator X preserving the pairing, $(1 \otimes X)K_0 = -(-1)^{|X|}(X \otimes 1)K_0$ up to the Koszul sign of Lemma c9-ops, so $(X \otimes 1 + 1 \otimes X)(P \star \otimes 1)K_0 = ([X, P\star] \otimes 1)K_0$ with the

graded commutator. (i) is eq c9-MP. (ii) is Prop c7-cov. (iii) and (iv) are Lemma c11-lines (ii), (iv): $[\eta., Q^{GF}]$ is obtained from $[x^\mu \xi_\mu ., Q^{GF}] = x^\mu C_{\xi_\mu} + [x^\mu, Q^{GF}]\xi_\mu .$, and for a function f, $[\eta., f(H)] = 2\sum_\mu \xi_\mu . \partial_\mu f'(H)$ because $\xi_\mu .$ and ∂_μ commute with H. (v) is Lemma c11-lines (v) and Duhamel's formula for $[v., e^{-tH}]$. □

Lemma 1.4 (Composite lines and their images under Q)**.** *A term of $S^{(1)}(x)$ with both legs on edges $P(\varepsilon, T)$ forms on a cycle the composite line along x, the kernel $(P \star x. \otimes 1 + 1 \otimes P \star x.)P(\varepsilon, T)$, built from $P \star x. P\star$.*

- *(i) It exists exactly for: the term $\int(x.A|_V, A^*|_V)$ of an even x acting nontrivially on $A|_V$ (Δ, P_b, M_ω, K_b), a line from c to $A|_V$; the term T_1 of $S^{(1)}(\xi)$, a line from λ to c; the terms $x^\mu T_1(\xi_\mu)$ of $S^{(1)}(\eta)$. Along R there is no composite line.*
- *(ii) For even x with $[x., Q] = 0$, $[Q, P \star x.P\star] = E(H)\, x.\, P \star - P \star x.\, E(H)$, which vanishes for Δ, P_b, M_ω: the composite line is Q-closed and its operator ∂ anticommutes with Q. Along K_b, $[Q, P \star v.P\star] = [v., P\star]E(H) - [v., E(H)]P\star$, nonzero in general.*
- *(iii) Along ξ, $[Q, P \star \xi.P\star] = E(H)C_\xi g(H) \neq 0$ (Lemma c11-lines (iii)).*

Proof. (i) $P(\varepsilon, T)$ has components in $A \otimes A$, $\lambda \otimes \lambda$, $c \otimes A^*|_V$, $A^*|_V \otimes c$ only (Prop c3-P), so a term of $S^{(1)}(x)$ with an antifield leg $A^*|_{W_\mathbb{R}^\perp}$, λ^* or c^* cannot have that leg on an edge P. For even x the only term with the antifield leg $A^*|_V$ is $\int(x.A|_V, A^*|_V)$; along R, $R.$ vanishes on $A|_V$ and c. For ξ it is T_1 (Table c11-terms). (ii) For even $X := P \star x.P\star$, by (1.1) and $[x., Q] = 0$,

$$Q\, P \star x.P\star = (E - P \star Q)x.P\star = Ex.P \star - P \star x.QP\star = Ex.P \star - P \star x.E + P \star x.P \star Q.$$

For P_b, M_ω, $x.$ commutes with Q^{GF} and H, hence with E and $P\star$, and the two terms cancel. For Δ this is Prop c9-composite (iii). For K_b, $Ex.P \star - P \star x.E = -[x., E]P \star + x.EP \star - P \star x.E = -[x., E]P \star + [x., P\star]E$, using $EP\star = P \star E$. (iii) is Lemma c11-lines (iii). □

1.2 The input from the action side

Hypothesis 1.5 (Action-side solution)**.** For a set $\mathfrak{X}$ of generators, $\hbar F_1 = \hbar F_1^{(0)} + \hbar F_1^{(1)} + \hbar F_1^{(2)}$ is a finite counterterm (Def c8-quant) that solves $s(\hbar F_1) = \hbar\, \mathrm{Ob}$ (Thm c8-obstruction (ii)) in arity zero and in arity one along every $x \in \mathfrak{X}$; in Sections **??** and in Theorem 3.2 (ii) also in arity two along the pairs considered there. In arity one along x the equation reads

$$(Q + \{I, -\})\, \hbar F_1^{(1)}(x) + (-1)^{|x|} \big\{ S^{(1)}(x), \hbar F_1^{(0)} \big\} = \hbar\, \mathrm{Ob}^{(1)}(x), \tag{1.4}$$

with $\mathrm{Ob}^{(1)}(x)$ the arity-one part of the obstruction of Def c8-ob along x, computed by the action workflow from the boundary and defect graphs of the action. $\hbar F_1^{(0)}$ contains the finite counterterm of Thm c4-finite and is invariant under translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_\mathbb{R}^\perp)$ and $\mathfrak{g}$, so that $\{S^{(1)}(x), \hbar F_1^{(0)}\} = 0$ for $x \in \{\Delta, P_b, M_\omega, R\}$. The solutions form an affine space over the s-closed local cochains of degree 0 (Thm c8-obstruction (iii)); the free directions that enter the one-body terms below are, in arity one:

x	s-closed directions of $\hbar F_1^{(1)}(x)$ that change the one-body kernel
Δ	the three sector parts of $S^{(1)}(\Delta)$ with coefficients $\theta, \theta_\lambda, \theta_V$ units; $Q\int(\iota_x A\|_V, c^*)$, $Q\int(\rho(x)\lambda^*, \lambda^*)$ with κ_V, κ_λ units (completed to s-exact directions); the parity-odd class $\int(\lambda^*, \rho(e_1e_2e_3e_4)\lambda)$, excluded by reflection compatibility (Def c11-free, Prop c11-tc0)
P_b	the sector parts of $S^{(1)}(P_b)$ with $\theta_P, \theta_{P,\lambda}, \theta_{P,V}$ units; Q-exact directions (Def c11-free)
M_ω	the sector parts of $S^{(1)}(M_\omega)$ in the sectors of $A\|_{W_\mathbb{R}^\perp}$, λ and $(A\|_V, c)$, with coefficients $\theta_M, \theta_{M,\lambda}, \theta_{M,V}$ units; s-exact directions
R	the two sector parts $\int(R.A\|_{W_\mathbb{R}^\perp}, A^*\|_{W_\mathbb{R}^\perp})$ and $-\int(R.\lambda, \lambda^*)$ in the quadratic sector, whose cubic completion is $S^{(1)}(R)$ alone; s-exact directions
ξ	$\theta_V^\xi S_2^V(\xi) + \theta_\perp^\xi S_2^\perp(\xi)$ units, extended for $\theta_V^\xi = \theta_\perp^\xi$ by $S^{(1)}(\xi)$, the arity-one value of G_B (Prop c11-freexi)
η	$\sum_\mu \int x^\mu S_2^V(\rho(e_\mu)\eta)$ and $\sum_\mu \int x^\mu S_2^\perp(\rho(e_\mu)\eta) + \int s_0$ (Prop c11-freexi (iv))
K_b	the s-closed local functionals of degree -1 and Weyl weight 4, quadratic in the position, linear in b: the sector parts of $S^{(1)}(K_b)$ and s-exact directions

The coupling direction $S - S^{(2)}$ and the outer direction G_B (Def c8-B) are always present.

Remark 1.6 (What is input and what is derived)**.** The numbers of $\hbar F_1^{(1)}(x)$ (the particular solution and the values of the free coefficients fixed by the cubic part of (1.4) or by the arity-two equations) are computed by the parallel action workflow; on the operator side they enter only through the one-body term $-\{\hbar F_1^{(1)}(x), -\}$ of Theorem 2.2 and through the arity-two equations of Section **??**. Every statement below is stated as a function of these numbers. The structural facts about $\mathrm{Ob}^{(1)}(x)$ that the operator side needs are proved here (Lemma 2.6), as consequences of the same defect flow.

1.3 Defect data of a pair

Definition 1.7 (Defect cycles, finite special vertices and two-body kernels)**.** Let x be a generator, X_a, X_b letters of any slots, $0 < \varepsilon < T \le \infty$.

- (i) A defect cycle of (X_a, X_b) along x is a chain $X_a - e_0 - V_1 - e_1 - \cdots - V_n - e_n - X_b$ of Def c10-chain whose vertices are vertices of I and exactly one vertex V_s given by a term of $S^{(1)}(x)$, entered and left through two of its legs, and whose edges are components of $P(\varepsilon, T)$ and exactly one special edge $e_\sigma = K_\varepsilon - K_0$. The placements of V_s are: on a line (both legs on edges P: a composite line of Lemma 1.4); at the special edge (one leg on e_σ, the other on an edge P or on a letter); cubic (a term T_5 with two legs on the cycle, the third an output).

- (ii) $\mathcal{D}_x(X_a, X_b; \varepsilon, T)$ is the sum of the defect cycles of the pair, amputated, outputs open, read with the sign of Def c11-defect (ii).

- (iii) $\mathrm{ct}^{(1)}_{x,m}(M)$, $\mathrm{lc}^{(1)}_x(M)$ are the coefficients of ε^{-m} and $\log\varepsilon$ of the arity-one counterterm of M along x, the singular part of the cycles through two letters of M with $S^{(1)}(x)$ on the cycle and no special edge; $\mathcal{F}_x(X_a, X_b)$ is the finite special vertex of Def c11-defect (iii),

$$\mathcal{F}_x(X_a, X_b) = -\sum_{m\ge 1} \frac{(-1)^m}{m!}\Big(\big\{I, \mathrm{ct}^{(1)}_{x,m}(X_aX_b)\big\}_{H^m} + \big\{S^{(1)}(x), \mathrm{ct}_m(X_aX_b)\big\}_{H^m}\Big),$$

  $\{-,-\}_{H^m}$ the bracket built from $(H^m \otimes 1)K_0$ (eq c10-finitesp).

- (iv) The two-body kernel is $\Upsilon_x(X_a, X_b) := [\varepsilon^0]\mathcal{D}_x(X_a, X_b; \varepsilon, T) + \mathcal{F}_x(X_a, X_b)$.
- (v) In arity zero, the special-edge cycles of the pair are the chains with vertices of I, edges $P(\varepsilon, T)$ and exactly one special edge, and $\Upsilon_0(X_a, X_b)$ is the coefficient of ε^0 of their sum plus $-\sum_{m\geq 1} \frac{(-1)^m}{m!}\{I, \mathrm{ct}_m(X_a X_b)\}_{H^m}$ (eq c10-finitesp, pair part).

For x even and acting on $A|_V$ the defect cycles with V_s on a line are the special-edge composite cycles of Def c11-sp; their coefficient of ε^0 is the composite pair kernel, written $\mathcal{C}_x(X_a, X_b)$.

Proposition 1.8 (Placements)**.** *The placements of the terms of $S^{(1)}(x)$ in the defect cycles are:*

x	*on a line*	*at the special edge*	*cubic*
Δ, P_b, M_ω, K_b	$\int(x.A|_V, A^*|_V)$ $(c \to A|_V)$	*all five slot pairs*	*none*
R	*none*	$(A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$, (λ, λ^*)	*none*
ξ	T_1 $(\lambda \to c)$	$T_1, \dots, T_4$	T_5 *(at e_σ or with an output λ^*)*
η	$x^\mu T_1(\xi_\mu)$	$x^\mu T_{1..4}(\xi_\mu)$, s_0, Y_η	$x^\mu T_5(\xi_\mu)$

Every letter is active along every generator except along R, where the letters c, $A|_V$, c^*, $A^*|_V$ meet $S^{(1)}(R)$ only through vertices of I.

Proof. Lemma 1.4 (i) for the lines; at the special edge one leg of the quadratic term is contracted through $K_\varepsilon - K_0$, which has all field–antifield components, and the other through P or directly with a letter; the cubic terms of $S^{(1)}(\xi)$ are the T_5 (Table c11-terms) and $S^{(1)}(x)$ is quadratic for even x. For R, $R.$ acts on $A|_{W_\mathbb{R}^\perp}$, λ and their antifields only. Activity: Lemma c10-active1. □

2 The components from the defect flow

2.1 The defect vertex of an operator

Let $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - \hbar F_1$ (Hypothesis 1.5), O a local functional of degree $|O|$, δ a square-zero parameter of degree $-|O|$, and $O(\varepsilon) = O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)$ (Def c8-renorm with $\psi = 0$). The equivariant counterterm of O has the arity-zero part $\partial_\delta I^{CT}[\varepsilon](O)$ and the arity-one part along x

$$\mathrm{ct}^{(1)}_x(O)[\varepsilon] = \sum_{m\geq 1} \varepsilon^{-m}\mathrm{ct}^{(1)}_{x,m}(O) + \log\varepsilon\,\mathrm{lc}^{(1)}_x(O).$$

For two functionals write $\{f, g\}_\varepsilon = \{f, g\} + \{f, g\}^\sigma_\varepsilon$; by $\Delta_\varepsilon - \Delta_0 = -\partial_{K_\varepsilon - K_0}$, $\{f, g\}^\sigma_\varepsilon$ is minus f and g joined by one special edge, and on a tree at a momentum p it is $\sum_{j\geq 1} \frac{(-\varepsilon)^j}{j!}\{f, g\}_{H^j}$ (Prop c11-master, proof of Thm c11-classes (iii)).

Lemma 2.1 (The defect vertex)**.** *Modulo $\hbar^2$ and constants, the defect vertex $\mathfrak{d} := (d_{CE} + Q)(J + \delta O(\varepsilon)) + \frac{1}{2}\{J + \delta O(\varepsilon), J + \delta O(\varepsilon)\}_\varepsilon + \hbar\Delta_\varepsilon(J + \delta O(\varepsilon))$ has the following parts.*

- *(a) δ-independent: $\frac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\}^\sigma_\varepsilon - s\big(I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \hbar F_1\big) - \{I_{\mathfrak{psl}(4|4)}, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \hbar F_1\}^\sigma_\varepsilon + \hbar\Delta_\varepsilon I_{\mathfrak{psl}(4|4)}$: the defect vertex of the action.*
- *(b) δ-linear, arity zero: $\delta\big(sO + \{I, O\}^\sigma_\varepsilon + \hbar\Delta_\varepsilon O - s\,\partial_\delta I^{CT}[\varepsilon](O) - \{I, \partial_\delta I^{CT}[\varepsilon](O)\}^\sigma_\varepsilon - \{I^{CT}[\varepsilon] + \hbar F^{(0)}_1, O\}_\varepsilon\big)$.*

- *(c) δ-linear, arity one along x: δ times*

$$\pm\{S^{(1)}(x),O\}\pm\{S^{(1)}(x),O\}^{\sigma}_{\varepsilon}-s\,\mathrm{ct}^{(1)}_{x}(O)[\varepsilon]-\{I,\mathrm{ct}^{(1)}_{x}(O)[\varepsilon]\}^{\sigma}_{\varepsilon}\mp\{S^{(1)}(x),\partial_{\delta}I^{CT}[\varepsilon](O)\}_{\varepsilon}$$
$$-\big\{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)+\hbar F^{(1)}_{1}(x),O\big\}_{\varepsilon},$$

the signs $\pm$ those of the cochain bracket of Def c2-dce with an arity-zero argument (Thm c8-L (iv)); for $|O|=0$ they are $+$.

Proof. Expand in δ, in the arity and in $\hbar$, with $I_{\mathfrak{psl}(4|4)}=I+S^{(1)}+S^{(2)}$ and $\partial_{\delta}I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)=\partial_{\delta}I^{CT}[\varepsilon](O)+\mathrm{ct}^{(1)}(O)[\varepsilon]+\cdots$. The classical master equation eq c2-mc, $(d_{CE}+Q)I_{\mathfrak{psl}(4|4)}+\frac{1}{2}\{I_{\mathfrak{psl}(4|4)},I_{\mathfrak{psl}(4|4)}\}=0$, removes the bracket without special edge in (a); d_{CE} vanishes on arity zero, so the arity-zero and arity-one parts of $(d_{CE}+Q)(\delta O)+\{I_{\mathfrak{psl}(4|4)},\delta O\}$ are δsO and $\pm\delta\{S^{(1)}(x),O\}$. The terms $\hbar\Delta_{\varepsilon}$ of the arity-one counterterms and of $\hbar F_1$ are of order $\hbar^2$, and so are the brackets of two counterterms. $\hbar\Delta_{\varepsilon}S^{(1)}(x)=0$: its quadratic terms pair a field with the antifield of another field, its cubic terms carry λ^* and no λ (proof of Thm c11-classes (i)). □

2.2 The master formula

Theorem 2.2 (Every component from the defect flow)**.** *Let x be a generator, $O=\int\omega\,X_1\cdots X_n$ a monomial local functional of degree 0, gauge-variant ones included, $\psi=0$, and assume Hypothesis 1.5 with $x\in\mathfrak{X}$. Take $T=\infty$ for $x\neq K_b$ and any $T\in(0,\infty)$ for $x=K_b$. Write the left side of eq c8-rep in arity one along x as $\big(L(O)^{(1)}(x)\big)[T]+\big(L(O)^{(0)}\big)[T]^{(1)}(x)$ with $L(O)^{(1)}(x)=\{S^{(1)}(x),O\}+\hbar L_1(O)^{(1)}(x)$, $L(O)^{(0)}=sO+\hbar L_1(O)^{(0)}$ (Thm c8-L (iv)).*

- *(i) (Classes.) The δ-linear graphs of arity one along x and of order $\hbar$ of the flow of $\mathfrak{d}$ with edges $P(\varepsilon,T)$ and other vertices from $J+\delta O(\varepsilon)$ form exactly the classes:*

  (F) one-loop graphs with the vertex $\pm\delta\{S^{(1)}(x),O\}$, and one-loop graphs with the vertex $\delta\,sO$ and a vertex $S^{(1)}(x)$ of J;

  (C2) graphs whose cycle passes through two letters of O and carries exactly one special edge and exactly one term of $S^{(1)}(x)$, with trees of I on the outputs: the defect cycles of Def 1.7 with trees;

  (C0) graphs whose cycle passes through two letters of O and carries one special edge or one contraction through K_{ε} ($\hbar\Delta_{\varepsilon}O$), the vertex $S^{(1)}(x)$ lying on a tree;

  (Br) graphs with a special edge on a tree;

  (Ac) graphs whose cycle passes through vertices of J alone, joined to O by a tree, together with the trees of the defect vertex (a) of the action;

  (Ct) trees with one vertex of order $\hbar$ from (b) or (c): the counterterm vertices, the finite special vertices produced by $\{\,,\}^{\sigma}_{\varepsilon}$ on the poles, and $-\{\hbar F_1,O\}_{\varepsilon}$.

- *(ii) As $\varepsilon\to 0$: (F) with the counterterm trees of $\{S^{(1)}(x),O\}$ and of sO equals the one-loop parts of $\big(\{S^{(1)}(x),O\}\big)[T]$ and of $(sO)[T]^{(1)}(x)$ on the left side; (C0) with the arity-zero vertices of (Ct) tends to $\big(\hbar L_1(O)^{(0)}\big)[T]^{(1)}(x)$, the arity-zero component transported by the trees with one vertex $S^{(1)}(x)$; (Br) tends to 0; (Ac) tends to 0 by (1.4) and the arity-zero equation.*

- *(iii) (Master formula.) With the pair sum (1.3),*

$$\hbar L_1(O)^{(1)}(x)=\sum_{a<b}\pm\Upsilon_x(X_a,X_b)-\big\{\hbar F^{(1)}_1(x),O\big\},\tag{2.1}$$

and in arity zero

$$\hbar L_1(O)^{(0)} = \sum_{a<b} \pm \Upsilon_0(X_a, X_b) - \{\hbar F_1^{(0)}, O\} - \sum_{m\geq 1} \frac{(-1)^m}{m!} \{\mathrm{ct}_m(I^{CT}), O\}_{H^m}, \tag{2.2}$$

$\mathrm{ct}_m(I^{CT})$ *the coefficient of* ε^{-m} *in* $I^{CT}[\varepsilon]$.

- *(iv) (Cancellations.) The defect cycles contain no* $\log\varepsilon$*; the coefficient of* $\log\varepsilon$ *cancels by the logarithmic covariance eq c11-logcov and the coefficients of* ε^{-m} *by the pole identity eq c11-pole;* $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ *has no pole along every generator.*
- *(v) (Endpoint.) Along* K_b *the right side of* (2.1) *is the same local functional at every finite* T*, and so is* $K_1(O)$*; along the other generators it is the same at every* $T \in (0,\infty]$.

Proof. By Lemma c8-transfer, $(d_{CE} + Q + \hbar\Delta_T) e^{\hbar\partial_{P(\varepsilon,T)}} e^{(J+\delta O(\varepsilon))/\hbar} = e^{\hbar\partial_{P(\varepsilon,T)}} \big(\hbar^{-1}\mathfrak{d}\, e^{(J+\delta O(\varepsilon))/\hbar}\big)$, so the left side of eq c8-rep at scale T, $d_T O[T]$, is the δ-linear part of the $\varepsilon\to 0$ limit of the connected graphs with one vertex $\mathfrak{d}$, other vertices from $J+\delta O(\varepsilon)$ and edges $P(\varepsilon,T)$ (Lemma c8-flow, Prop c4-defectflow), and by Def c8-rep it equals $L(O)[T]$.

(i) At order $\hbar$ a connected graph has one cycle and vertices of order $\hbar^0$, or no cycle and one vertex of order $\hbar$ (Prop c3-graphs). Being δ-linear and of arity one along x, exactly one vertex carries δ and exactly one carries x, possibly the same. The vertex $\mathfrak{d}$ is one of (a), (b), (c) of Lemma 2.1. A cycle passes through O along exactly two letters or avoids O (Thm c10-decomp (i)); in the first case the positions of the special edge and of the term of $S^{(1)}(x)$ relative to the cycle sort the graph into (C2), (C0) or (Br), the vertex $\pm\delta\{S^{(1)}(x), O\}$ and the vertex δsO into (F); in the second case into (Ac) when the special edge or the action defect lies on the cycle, into (Br) when it lies on a tree. Trees with a vertex of order $\hbar$ are (Ct). The list is exhaustive, every vertex of (a)–(c) having been placed.

(ii) (F): these are, graph by graph, the one-loop graphs of the observables $\{S^{(1)}(x), O\}$ and sO at scale T with $S^{(1)}(x)$ on the cycle or on a tree; the counterterm trees $-\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x), O\})$ and $-\partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](sO)$ are not vertices of $\mathfrak{d}$, and their differences with the vertices $\mp\{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\}$ and $-s\,\partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)$ of (b), (c) are local and singular; they are kept in (Ct) and enter (iv). (C0): the cycle is a special-edge cycle or a K_ε-contraction through two letters of O, with $S^{(1)}(x)$ on a tree attached at generic momenta; by Thm c10-brst (proof), eq c10-finitesp and the arity-zero part of the present argument (which is (iii) at arity zero) the cycles with their arity-zero counterterm vertices converge to the trees of $\hbar L_1(O)^{(0)}$, here with one tree vertex $S^{(1)}(x)$. (Br): a special edge on a tree at a generic momentum p multiplies a subgraph by $e^{-\varepsilon|p|^2}-1$; the subgraph together with its counterterm trees has a limit, and the counterterm vertices with the same special edge are of the form $(e^{-\varepsilon|p|^2}-1)\varepsilon^{-m}$ or $(e^{-\varepsilon|p|^2}-1)\log\varepsilon$ times a local vertex: the first are pure powers ε^{j-m} times H^j, which are accounted for in (Ct) as finite special vertices and poles, the second tend to 0; what remains is $e^{-\varepsilon|p|^2}-1$ times a convergent graph and tends to 0 (Thm c8-obstruction (ii), Prop c4-defect1). Along K_b at finite T the subgraphs carry at fixed ε the term $\frac{3}{8\pi^2}\hbar\log T$ times a tree with the same factor (Rem after Thm c8-K), which also tends to 0. (Ac): a cycle through vertices of J alone, with its counterterm and finite vertices of the action, is a graph of the flow of the defect vertex (a) of the action, attached to O by a tree; its $\varepsilon\to 0$ limit is the tree-level flow of $\hbar\,\mathrm{Ob} - s(\hbar F_1)$ (Def c8-ob, Thm c8-obstruction (ii)), whose arity-zero part and arity-one part along x vanish by Hypothesis 1.5; the tree joining it to O has generic momenta.

(iii) What remains is (C2) and the arity-one vertices of (Ct): $-s\,\mathrm{ct}^{(1)}_x(O)[\varepsilon]$, $-\{I, \mathrm{ct}^{(1)}_x(O)[\varepsilon]\}^\sigma_\varepsilon$, $\mp\{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon$, $-\{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O\}_\varepsilon$, $-\{\hbar F_1^{(1)}(x), O\}_\varepsilon$, together with the counterterm trees $-\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x), O\})$ and $-\mathrm{ct}^{(1)}_x(sO)$ moved from the left side. Each defect cycle of a pair is, by Thm c11-oddsimplex (i) at $T=\infty$ and (iii) at finite T, $\sum_{m\geq 1}\varepsilon^{-m}a_m + a_0 + o(1)$ with local coefficients and no $\log\varepsilon$; the trees on its outputs have generic momenta and converge to trees with edges $P(0,T)$.

In a special-edge part $-\varepsilon^{-m}\{f,g_m\}^\sigma_\varepsilon$ of a counterterm bracket the term of order ε^0 is $-\frac{(-1)^m}{m!}\{f,g_m\}_{H^m}$, the terms with $j<m$ are poles and those with $j>m$ tend to 0; $\log\varepsilon$ times a special edge tends to 0. For $f=I$, $g_m=\mathrm{ct}^{(1)}_{x,m}(O)$ and for $f=S^{(1)}(x)$, $g_m=\mathrm{ct}_m(O)$ these are the finite special vertices, two-body because ct_m, $\mathrm{ct}^{(1)}_{x,m}$ are sums over pairs (Thm c10-decomp (iii) and the same factorisation of the cycles through two letters with $S^{(1)}(x)$ on the cycle); for $f=I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ there is no pole by (iv). The term $-\{\hbar F_1^{(1)}(x),O\}_\varepsilon$ gives $-\{\hbar F_1^{(1)}(x),O\}$, its special-edge part tending to 0. After the cancellations of (iv) the sum converges to the tree flow of the right side of (2.1), and Thm c8-L (i) identifies the local functionals. The sum over pairs with the Koszul signs is that of Thm c10-decomp (iii), each defect cycle meeting O in exactly two letters. At arity zero the same argument with the vertices (b) gives (2.2); the last term is the finite special vertex of $\{I^{CT}[\varepsilon],O\}^\sigma_\varepsilon$, one-body.

(iv) The left side has a limit, so the coefficients of $\log\varepsilon$ and of ε^{-m} of the collected vertices vanish; with the injectivity of the tree flow (Thm c8-L (i)) these are eq c11-logcov and eq c11-pole (Thm c11-classes (iv)). A coefficient of ε^{-m} of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is a colour-singlet local functional of degree -1 and total Weyl weight $4-\frac12\lambda_x-2m$, the position at weight -1 and of degree p_x at most that of x (Prop c10-bound1); a colour singlet of degree -1 has an antifield of weight at least $\frac52$ and a second adjoint field of weight at least 1, or c^* and c, so its total weight is at least $\frac72-p_x$; for $m\geq1$ the total weight is at most $2,3,2,2,\frac52,\frac32,1$ along $\Delta,P_b,M_\omega,R,\Sigma,\Sigma^*,K_b$, below $\frac52,\frac72,\frac52,\frac72,\frac72,\frac52,\frac32$.

(v) Thm c11-oddsimplex (iii): at finite T the coefficients of the expansion in ε are those of the region of small total Schwinger time, and $T\partial_T$ of every defect cycle is $O(\varepsilon)$, because the edge pinned at T confines the loop momentum while the special edge carries $e^{-\varepsilon|l+q_\sigma|^2}-1=O(\varepsilon(1+|l|)^2)$; the finite special vertices are built from ct_m, $\mathrm{ct}^{(1)}_{x,m}$, independent of T (Prop c8-Tindep); $\hbar F_1^{(1)}(K_b)$ is independent of T. Along the other generators Prop c8-Tsafe and Thm c8-L (ii). □

Remark 2.3 (What the master formula uses)**.** The proof uses Lemma c8-transfer, Def c8-rep and the injectivity of the tree flow, the classical master equation, the action-side equations (1.4) through arity one along x, the simplex form of the defect cycles (no $\log\varepsilon$ on a cycle with a special edge, a property of the cutoff on every edge), and generic momenta at $T=\infty$ or any finite T. It uses neither the scaling of the regulator, nor the composite term, nor any relation between generators: every number of (2.1) is a coefficient of ε^0 of a defect cycle, a finite special vertex, or a coefficient of the action-side solution. Equation (2.1) is therefore the definition of the numbers of the components in the strict route; the theorems below are identities proved from it.

2.3 The Ward identity of the flow along an even generator

Theorem 2.4 (Ward identity of the flow)**.** *Let x be an even generator, J even and at least cubic modulo $\hbar$, $P=P(\varepsilon,T)$, $W=W(P,J)$. Then, with $\{S^{(1)}(x),-\}=x.$ on functionals (with the sign of Def c7-vdot),*

$$\{S^{(1)}(x),W\}=\partial_\delta W\big(P,J+\delta\{S^{(1)}(x),J\}\big)-\hbar\,\partial_{\mathcal{V}_x}W-\tfrac12\big(\partial_{\mathcal{V}_x}(W^2)-2W\,\partial_{\mathcal{V}_x}W\big),\tag{2.3}$$

the last two terms being the graphs of W with one edge P replaced by $\mathcal{V}_x(\varepsilon,T)$. In particular:

- *(i) for $x=\Delta$, $\{S^{(1)}(\Delta),W\}=\partial_\delta W(P,J+\delta\{S^{(1)}(\Delta),J\})+4(\varepsilon\partial_\varepsilon+T\partial_T)W$, the derivatives acting on P (eq c9-ward);*
- *(ii) for $x\in\{P_b,M_\omega,R\}$, $\{S^{(1)}(x),W\}=\partial_\delta W(P,J+\delta\{S^{(1)}(x),J\})$ exactly at every ε and T (eq c7-flowcov).*

Proof. By Lemma c9-ops, $[\partial_\phi,x.]=\partial_{(x.\otimes1+1\otimes x.)\phi}$ for even $x.$, so $x.\,\partial_P=\partial_P\,x.-\partial_{\mathcal{V}_x}$; the operators ∂_ϕ commute, hence $x.\,e^{\hbar\partial_P}=e^{\hbar\partial_P}\big(x.-\hbar\,\partial_{\mathcal{V}_x}\big)$. Apply to $e^{J/\hbar}$: $x.\,e^{W/\hbar}=\hbar^{-1}(x.W)e^{W/\hbar}$, $e^{\hbar\partial_P}(x.e^{J/\hbar})=$

$\hbar^{-1}\partial_\delta W(P, J+\delta\, x.J)\, e^{W/\hbar}$, and $\partial_{\mathcal{V}_x} e^{W/\hbar} = \big(\hbar^{-1}\partial_{\mathcal{V}_x} W + \frac{1}{2}\hbar^{-2}(\partial_{\mathcal{V}_x}(W^2) - 2W\partial_{\mathcal{V}_x} W)\big)e^{W/\hbar}$ by the product rule of the second-order operator (Def c3-delta); multiply by $\hbar\, e^{-W/\hbar}$. (i) by Lemma 1.3 (i): an edge replaced by $-4(\varepsilon\partial_\varepsilon + T\partial_T)P$, summed over the edges, is $-4(\varepsilon\partial_\varepsilon + T\partial_T)$ of the graph. (ii) by Lemma 1.3 (ii). □

Proposition 2.5 (Graphs with one even vertex and their images under Q)**.** *Let x be even, $P = P(\varepsilon, T)$, J as in Theorem 2.4. As operators on functionals*

$$e^{\hbar\partial_P} S^{(1)}(x) = \Big(S^{(1)}(x) + \hbar\, P\star\, x. + \tfrac{\hbar^2}{2}\,\partial_{\mathsf{L}_x}\Big) e^{\hbar\partial_P} + \hbar^2 c_x,$$

with $\mathsf{L}_x := (P\star x.\otimes 1 + 1\otimes P\star x.)P$ the composite line kernel and c_x a constant, so the linearised flow of $S^{(1)}(x)$ is the vertex alone, the vertex joined by one edge to a leg of W, two legs of W closed by the composite line, and two copies of W joined by it. If L_x is Q-closed (Lemma 1.4 (ii): $x \in \{\Delta, P_b, M_\omega\}$, and $x = R$, where $\mathsf{L}_R = 0$), then

$$Q\Big(e^{-W/\hbar} e^{\hbar\partial_P}\big(S^{(1)}(x) e^{J/\hbar}\big)\Big) = \{S^{(1)}(x), W\}_\varepsilon - \{S^{(1)}(x), W\}_T - \hbar^{-1} e^{-W/\hbar}\Big[S^{(1)}(x) + \hbar P\star x. + \tfrac{\hbar^2}{2}\partial_{\mathsf{L}_x},\, QW\Big] e^{W/\hbar}.$$

Proof. As Prop c9-conj and Prop c9-Qconj with $x.$ in place of $v.$: $\mathrm{ad}_{\partial_P} S^{(1)}(x) = P\star x.$ plus a constant (the full contraction of the quadratic $S^{(1)}(x)$ with P, zero unless P has a component in the slot pair of a term of $S^{(1)}(x)$, which happens only for $\int(x.A|_V, A^*|_V)$ and the component $A\otimes A$, $c\otimes A^*|_V$ in different slots, so the constant is a trace that does not enter connected graphs with legs), $\mathrm{ad}_{\partial_P}(P\star x.) = \partial_{\mathsf{L}_x}$ by Lemma c9-ops, and the series stops. For the image, $QP\star x. + P\star x.Q = E(H)x.$ by (1.1) and $[x., Q] = 0$, $\{S^{(1)}(x), F\}_t = (e^{-tH}x.)F$ (Lemma c9-SDelta (ii), whose proof uses only that $S^{(1)}(x)$ is quadratic with the arrows of $x.$), and $Q\, S^{(1)}(x) = 0$ for even x ($[x., Q] = 0$ and preservation of the pairing). □

Lemma 2.6 (The arity-one obstruction along an even generator with a closed line)**.** *For $x \in \{\Delta, P_b, M_\omega, R\}$, assume the arity-zero part of Hypothesis 1.5 and that $\hbar F_1^{(0)}$ is invariant under x. Then $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is the singular part of the composite loops of I, $\frac{\hbar}{2}\partial_{\mathsf{L}_x} W(P(\varepsilon,\infty), I)_{\mathrm{tree}}$, and*

$$\hbar\,\mathrm{Ob}^{(1)}(x) = \mathcal{V}_x^{\mathrm{act}} + (\textit{composite part along } x), \qquad \mathcal{V}_\Delta^{\mathrm{act}} = 4\, s(\textit{unit}\cdot\mathcal{N}), \quad \mathcal{V}_{P_b}^{\mathrm{act}} = \mathcal{V}_{M_\omega}^{\mathrm{act}} = \mathcal{V}_R^{\mathrm{act}} = 0,$$

where the composite part along x is the local functional whose tree flow is $\lim_{\varepsilon\to 0}\big(Q(\textit{composite loops of } I) - \textit{tree flow of } s\, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)\big)$; by Lemma 4.2 it is $[\varepsilon^0]\Xi_x^\sigma$, the coefficient of ε^0 of the amputated composite loops of I with one special edge on the cycle, and it is 0 for $x = R$. In particular $\mathrm{Ob}^{(1)}(R) = 0$ and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(R) = 0$.

Proof. As Prop c9-structure, with Proposition 2.5 and Theorem 2.4 in place of eq c9-D1a, eq c9-D1b and eq c9-ward: at $T = \infty$, Q of the linearised flow of $S^{(1)}(x)$ at $J = I - I^{CT}[\varepsilon] - \hbar F_1^{(0)}$ is $\{S^{(1)}(x), W\}_\varepsilon$ minus the terms with QW; the special-edge part of $\{S^{(1)}(x), W\}_\varepsilon$ on legs tends to 0; by (2.3), $\{S^{(1)}(x), W\}$ is the linearised flow of $x.J$ plus the variation term. For Δ, $x.I = x.\hbar F_1^{(0)} = 0$, $(v. - 4\varepsilon\frac{d}{d\varepsilon})I^{CT}[\varepsilon] = -4$ times the coefficient of $\log\varepsilon$ (proof of Cor c9-CS), and $4\varepsilon\partial_\varepsilon W \to 0$; this gives $\mathcal{V}_\Delta^{\mathrm{act}}$. For P_b, M_ω, R, $x.I = 0$, $x.I^{CT}[\varepsilon] = 0$ by the covariance of the counterterms (Prop c10-map (iv)), $x.\hbar F_1^{(0)} = 0$ by assumption, and there is no variation term. The terms with QW tend to 0 by the arity-zero equation, except $-\frac{\hbar}{2}\partial_{\mathsf{L}_x}(QW_{\mathrm{tree}}) = Q(\frac{\hbar}{2}\partial_{\mathsf{L}_x} W_{\mathrm{tree}})$, Q of the composite loops. The graphs with $S^{(1)}(x)$ on an edge at a leg of the one-loop part of W converge, so the singular part is that of the composite loops. For R there is no composite line (Lemma 1.4 (i)), so the counterterm and the composite part vanish. □

2.4 The second evaluation and the identities along the even generators

Definition 2.7 (Composite term)**.** For $x \in \{\Delta, P_b, M_\omega\}$ and a local functional O, the composite graphs of O along x are the δ-linear part of $\frac{\hbar}{2}\partial_{\mathsf{L}_x} W(P, I+\delta O)_{\mathrm{tree}}$ (Def c11-graphs), their counterterm trees are those of $\mathrm{ct}_x^{(1)}(O)$ and of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, and the composite term of O along x is the local functional, when it exists, whose tree flow at scale T is

$$\lim_{\varepsilon\to 0} Q\big(\text{composite graphs of } O - \text{their counterterm trees}\big)[T] - \big(\text{tree insertion of the composite part along } x \text{ into } O[T]\big)$$

(Def c10-hyp for Δ). Along R the composite term is 0.

Theorem 2.8 (The conjugation evaluation along Δ, P_b, M_ω, R)**.** *Let $x \in \{\Delta, P_b, M_\omega, R\}$, assume Hypothesis 1.5 along x with $\hbar F_1^{(0)}$ invariant under x, and let O be homogeneous with $sO = 0$ and $\hbar L_1(O)^{(0)} = 0$ for a finite part ψ of the operators that is linear in ω, translation invariant, weight preserving and covariant under $\mathfrak{so}(V) \oplus \mathfrak{so}(W^\perp_{\mathbb{R}})$. Then at every $T \in (0,\infty]$*

$$\hbar L_1(O)^{(1)}(x) = \mathcal{V}_x(O) + (\textit{composite term of } O \textit{ along } x) - \big\{\hbar F_1^{(1)}(x), O\big\}, \qquad \mathcal{V}_\Delta(O) = 4\mathrm{lc}(O), \quad \mathcal{V}_{P_b} = \mathcal{V}_{M_\omega} = \mathcal{V}_R = 0 \tag{2.4}$$

Proof. At $T=\infty$, $d_\infty = d_{CE} + Q$, and the arity-one component along x of $d_\infty O[\infty]$ is Q of the arity-one part of $O[\infty]$, which is the $\varepsilon \to 0$ limit of the δ-part of the linearised flow at $J + \delta O(\varepsilon)$ of $S^{(1)}(x) - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) - \hbar F_1^{(1)}(x) - \delta\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]$. With $sO = 0$ and $L_1(O)^{(0)} = 0$ the left side of eq c8-rep in arity one is $\big(\{S^{(1)}(x), O\} + \hbar L_1(O)^{(1)}(x)\big)[\infty]$. (1) The vertex $S^{(1)}(x)$: by Proposition 2.5, Q of its linearised flow is $\{S^{(1)}(x), W\}_\varepsilon$ minus the terms with QW. The special-edge part on legs tends to 0. By Theorem 2.4, $\{S^{(1)}(x), W\}$ is the linearised flow of $x.J$ plus the variation term. The δ-part of $x.J$ is $x.O(\varepsilon)$: for $x = \Delta$, $(v. - 4\varepsilon\frac{d}{d\varepsilon})O(\varepsilon) = (v.O)(\varepsilon) + 4\mathrm{lc}(O)$ by eq c10-scaling, the variation term $4\varepsilon\partial_\varepsilon W$ tends to 0 after renormalisation (proof of Cor c9-CS), and the δ-independent part gives 4 times the tree insertion of the coefficient of $\log\varepsilon$ of $I^{CT}[\varepsilon]$ into $O[\infty]$; for $x \in \{P_b, M_\omega, R\}$, $x.\partial_\delta I^{CT}[\varepsilon](O) = \partial_\delta I^{CT}[\varepsilon](x.O)$ by the covariance of the counterterm map (Prop c10-map (iv)) and $x.\psi = \psi x.$, so the δ-part tends to $(x.O)[\infty]$ and there is no variation term. In the terms with QW, QW is the flow of the arity-zero defect vertex; its δ-independent limit vanishes by the arity-zero equation and its δ-part by $sO = 0$ and $L_1(O)^{(0)} = 0$, except Q of the composite graphs, $-\frac{\hbar}{2}\partial_{\mathsf{L}_x}(QW_{\mathrm{tree}}) = Q(\frac{\hbar}{2}\partial_{\mathsf{L}_x} W_{\mathrm{tree}})$, by Lemma 1.4 (ii). (2) The vertex $-\hbar F_1^{(1)}(x)$ enters by tree insertion; by Lemma c6-bvexact with $sO = 0$, $-Q$ of its tree insertion is minus the tree insertion of $s\hbar F_1^{(1)}(x) = \hbar \mathrm{Ob}^{(1)}(x)$, which by Lemma 2.6 is $\mathcal{V}_x^{\mathrm{act}}$ plus the composite part, minus $\{\hbar F_1^{(1)}(x), O\}[\infty]$; the first cancels the δ-independent term of (1) along Δ. (3) The vertices $-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ and $-\delta\mathrm{ct}_x^{(1)}(O)$ are the counterterm trees of the composite graphs, and Q of them completes Q of the composite graphs to Q of the renormalised ones. Collecting, with $\{S^{(1)}(x), O\} = x.O$, the arity-one component is $(x.O)[\infty] + \mathcal{V}_x(O)[\infty] + (\text{composite term})[\infty] - \{\hbar F_1^{(1)}(x), O\}[\infty]$, and Thm c8-L (i) gives (2.4). Along R the composite graphs and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(R)$ vanish. By Thm c8-L (ii) the same local functional is the representative at every finite T. □

Theorem 2.9 (Dilatation, translations and rotations from the defect flow)**.** *Under the hypotheses of Theorem 2.8 and of the locality Theorem 4.5 (iv), for every cocycle $O = \int \omega\, X_1 \cdots X_n$:*

- *(a) $D_1(O) = 4\,\mathrm{lc}^{\mathrm{nat}}(O) + \sum_{a<b} \pm\mathcal{C}_\Delta(X_a, X_b) - \big\{\hbar F_1^{(1)}(\Delta) - 4\,\mathrm{unit}\cdot\mathcal{N}, O\big\}$: four times the natural logarithmic map, plus the composite pair kernels, plus a one-body term. Equivalently*

$$\sum_{a<b} \pm\Upsilon_\Delta(X_a, X_b) = \sum_{a<b} \pm\Big(4\mathrm{lc}(X_a X_b) + \mathcal{C}_\Delta(X_a, X_b)\Big). \tag{2.5}$$

- *(b)* $P_1(O) = \sum_{a<b} \pm \mathcal{C}_{P_b}(X_a, X_b) - \{\hbar F_1^{(1)}(P_b), O\}$ *and* $\sum_{a<b} \pm \Upsilon_{P_b} = \sum_{a<b} \pm \mathcal{C}_{P_b}$.
- *(c)* $M_1(O) = \sum_{a<b} \pm \mathcal{C}_{M_\omega}(X_a, X_b) - \{\hbar F_1^{(1)}(M_\omega), O\}$ *and* $\sum_{a<b} \pm \Upsilon_{M_\omega} = \sum_{a<b} \pm \mathcal{C}_{M_\omega}$.
- *(d)* $R_1(O) = -\{\hbar F_1^{(1)}(R), O\}$ *and* $\sum_{a<b} \pm \Upsilon_R(X_a, X_b) = 0$.

Proof. Theorem 2.2 (iii) and Theorem 2.8 compute the same local representative, which is unique (Thm c8-L (i)); the one-body terms $-\{\hbar F_1^{(1)}(x), O\}$ are the same in both. By Theorem 4.5 (iv) (with its extension to M_ω, Remark 4.6) the composite term of a cocycle is $\sum_{a<b} \pm \mathcal{C}_x(X_a, X_b)$. For (a), $4\mathrm{lc}(O) = \sum_{a<b} \pm 4\mathrm{lc}(X_a X_b)$ (Thm c10-decomp (iii)) and $4\mathrm{lc}(O) = 4\mathrm{lc}^{\mathrm{nat}}(O) + \{4\,\mathrm{unit} \cdot \mathcal{N}, O\}$ by (1.2). □

Remark 2.10 (Status of the identities)**.** Theorem 2.9 is an identity between two evaluations of the same defect flow (Lemma c8-transfer): the strict evaluation (2.1), which defines the numbers, and the evaluation through the closed form of the graphs with one even vertex and the regulator variation (Theorem 2.4, a property of $P(\varepsilon, T)$ proved from its definition). The factor four and the logarithmic coefficient are outputs of the theorem; they enter no number. The theorem holds summed over the pairs of a cocycle; the corresponding statement for each pair is Conjecture 2.11, a computed identity of the engines (gate G-D1pair).

Conjecture 2.11 (Pair form)**.** *For every pair of letters of any slots, with any derivatives,*

$$\Upsilon_\Delta(X_a, X_b) = 4\mathrm{lc}(X_a X_b) + \mathcal{C}_\Delta(X_a, X_b), \qquad \Upsilon_{P_b}(X_a, X_b) = \mathcal{C}_{P_b}(X_a, X_b), \qquad \Upsilon_{M_\omega}(X_a, X_b) = \mathcal{C}_{M_\omega}(X_a, X_b), \qquad \Upsilon_R($$

Remark 2.12 (Evidence and mechanism)**.** (1) The pair kernels are graded symmetric, $\Upsilon_x(X_b, X_a) = \pm \Upsilon_x(X_a, X_b)$ with the Koszul sign, and so are lc and $\mathcal{C}_x$; the difference $\Upsilon_x - \mathcal{V}_x - \mathcal{C}_x$ is a two-body kernel whose sum over the pairs of every cocycle vanishes by Theorem 2.9. On the pair $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ with the outputs $A|_{W_{\mathbb{R}}^\perp} A|_{W_{\mathbb{R}}^\perp}$ the conjecture holds: the scalar bilinears are cocycles with this single pair, and Cor c11-DPdefect gives $\Upsilon_\Delta^0 = \frac{40}{3} O$ (traceless), $-\frac{104}{3} O$ (trace) units with $4\mathrm{lc} = 16O$, $-32O$ units and $\mathcal{C}_\Delta^0 = -\frac{8}{3} O$ units (Ex c11-bilinear), and $\Upsilon_{P_b} = \mathcal{C}_{P_b} = -\frac{1}{3}$ units times $\partial_b O$. Along R on this pair, graded symmetry and $\mathfrak{so}(W_{\mathbb{R}}^\perp)$-covariance leave the form $a\big((R.A|_{W_{\mathbb{R}}^\perp})A|_{W_{\mathbb{R}}^\perp} + A|_{W_{\mathbb{R}}^\perp}(R.A|_{W_{\mathbb{R}}^\perp})\big)$, which the cocycle identity on the traceless bilinears sets to 0. (2) The defect cycles along an even x with V_s at the special edge combine, on the edge e_σ, into $(K_\varepsilon - K_0) \star x.P \star + P \star x.(K_\varepsilon - K_0) \star = x.Y + Y x. + [(K_\varepsilon - K_0) \star, x.]P \star - P \star [(K_\varepsilon - K_0) \star, x.]$ with $Y = (K_\varepsilon - K_0) \star P \star = -P(\varepsilon, 2\varepsilon) \star$ at $T = \infty$, an edge with Schwinger time in $[\varepsilon, 2\varepsilon]$; the last two terms cancel for Δ, P_b, M_ω, R by Lemma 1.3. The conjecture states that these placements, with the finite special vertex, reproduce four times the logarithmic pair kernel along Δ and vanish along P_b, M_ω, R, pair by pair. (3) A proof pair by pair would follow from Theorem 2.8 for the non-cocycle $O = \int \omega\, X_a X_b$, which requires the terms with sO of both evaluations; by Theorem 4.5 (vi) these are the composite graphs of sO in the second and the classes (F), (C0) in the first.

3 Position decompositions and $K = P^\dagger$

3.1 Position decompositions

Definition 3.1 (Centred Killing data)**.** For a point x_O and $z := x - x_O$, the Killing data of the generators with explicit position decompose as

$$\Delta:\ 2x = 2x_O + 2z; \qquad M_\omega:\ \omega x = \omega x_O + \omega z; \qquad \eta:\ \rho(x)\eta = \rho(x_O)\eta + \rho(z)\eta;$$
$$K_b:\ v(x) = v(x_O) + \big(-2(b, x_O)z + \omega_{x_O} z\big) + v(z), \qquad \omega_{x_O} z := 2\big((x_O, z)b - (b, z)x_O\big).$$

For a generator y with Killing data of positive degree, $\Upsilon^0_y(X_a, X_b)$ denotes the two-body kernel at the point x_O of the pair computed with the Killing data of y centred at x_O, i.e. with x replaced by z; on plane waves, with the momentum r injected at V_s and absorbed by ω, the monomials z^μ, $z^\mu z^\nu$ of the Killing data act as $-i\partial_{r_\mu}$ and $-\partial_{r_\mu}\partial_{r_\nu}$ at $r=0$. Two-component: $\rho(z)\eta$ has the components $-i\,z_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $i\,z_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$ (eq c8-killing), and $(\omega_{x_O}z)^m = 2(x_O\cdot z)b^m - 2(b\cdot z)x_O^m$.

Theorem 3.2 (Position decompositions)**.** *Let the hypotheses of Theorem 2.2 hold along the generators involved.*

- *(i) (Two-body kernels, graph by graph.) For every pair of letters at the point x_O,*

$$\begin{aligned}&\Upsilon_\Delta = 2x_O^\mu\,\Upsilon_{P_{e_\mu}} + \Upsilon^0_\Delta, \qquad \Upsilon_{M_\omega} = (\omega x_O)^\mu\,\Upsilon_{P_{e_\mu}} + \Upsilon^0_{M_\omega}, \qquad \Upsilon_\eta = \sum_\mu x_O^\mu\,\Upsilon_{\rho(e_\mu)\eta} + \Upsilon^0_\eta,\\ &\Upsilon_{K_b} = v(x_O)^\mu\,\Upsilon_{P_{e_\mu}} - (b,x_O)\,\Upsilon^0_\Delta + \Upsilon^0_{M_{\omega_{x_O}}} + \Upsilon^0_{K_b},\end{aligned} \tag{3.1}$$

  the same decompositions holding separately for $[\varepsilon^0]\mathcal{D}_x$, for the finite special vertices $\mathcal{F}_x$, for every coefficient $\mathrm{ct}^{(1)}_{x,m}$, $\mathrm{lc}^{(1)}_x$ and for $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$. Υ^0_η is computed with $S^{(1)}(\eta)$ at the spinor $\rho(z)\eta$ together with the term s_0; along η the term Y_η of η. belongs to Υ^0_η.

- *(ii) (One-body terms.) Let $\hbar F_1$ solve the arity-two part of $s(\hbar F_1) = \hbar\mathrm{Ob}$ along (y,P_b) for $y\in\{\Delta, M_\omega, \eta, K_b\}$. Then, with $S^{(2)}(y,P_b)=0$ (Def c2-S2) and the sign rule of Def c2-dce,*

$$\big(\text{position part along } b \text{ of } \hbar F_1^{(1)}(y)\big) = \pm\hbar F_1^{(1)}([y,P_b]) \pm \big\{S^{(1)}(y), \hbar F_1^{(1)}(P_b)\big\} \pm s\big(\hbar F_1^{(2)}(y,P_b)\big) \mp \hbar\,\mathrm{Ob}^{(2)}(y,P_b), \tag{3.2}$$

  with $[\Delta,P_b] = -2P_b$, $[M_\omega,P_b] = -P_{\omega b}$, $[\eta,P_b] = -\rho(b)\eta$, $[K_b,P_{b'}] = (b,b')\Delta + M_{bb'}$ (Rem c7-KDelta). In a scheme in which $\hbar F_1^{(1)}(P_b)=0$ and the arity-two obstruction along (y,P_b) is s of a local functional, the one-body term $-\{\hbar F_1^{(1)}(y), O\}$ decomposes as (3.1) up to s-exact brackets, which are zero on cohomology (Lemma c10-deriv).

- *(iii) (Components on cohomology.) If the local representatives exist through arity two along (y,P_b) on a family stable under Λ (Cor c8-homotopy), then on its cocycles, with $\Lambda^{(1)}(x) = \Lambda^{(1)}_0(x) + \hbar\Lambda^{(1)}_1(x)$ as in Prop c11-relations,*

$$\big[\Lambda^{(1)}_1(y), \Lambda^{(1)}_0(P_b)\big] + \big[\Lambda^{(1)}_0(y), \Lambda^{(1)}_1(P_b)\big] + \Lambda^{(1)}_1([y,P_b]) \cong 0 \tag{3.3}$$

  modulo $\Lambda^{(0)}$-exact terms; by Lemma c11-canonical the first term is the position part of the component along y, up to the sign fixed by the order-$\hbar^0$ identity. In particular: the position part of D_1 vanishes on cohomology when P_1 is translation covariant (Thm c11-arity2); the position part of S_1 along η is, on cohomology, Q_1 along $\rho(b)\eta$ up to the commutator of $\{S^{(1)}(\eta),-\}$ with P_1 (Prop c11-S (ii)); the position part of K_1 along b' is, on cohomology, the component along $(b,b')\Delta + M_{bb'}$ up to the commutator of $\{S^{(1)}(K_b),-\}$ with P_1 (Prop c11-relations (R5)).

Proof. (i) $S^{(1)}(y)$ is linear in the Killing data of y: for even y through $v_\cdot = v^\mu\partial_\mu + \frac12\omega_{\mu\nu}\Sigma^{\mu\nu} + \frac{w}{4}\operatorname{div} v$ (proof of Lemma c11-lines (v)), whose three terms are linear in v, dv, $\operatorname{div} v$; for η through eq c11-Seta, $S^{(1)}(\eta) = \sum_\mu\int x^\mu s_\mu + \int s_0$ (Table c11-terms, Lemma c11-lines (iv)). Insert the decomposition of Def 3.1. On plane waves the factor $x^\mu = x_O^\mu + z^\mu$ at V_s acts as x_O^μ on the delta function of the total momentum absorbed by ω (the point x_O of the density) and as $-i\partial_{r_\mu}$ on the integrand (Prop c11-position (i)); the constant part $v(x_O)$ of the vector field of K_b is a translation, its linear part $-2(b,x_O)z + \omega_{x_O}z$ is $-(b,x_O)$ times the dilatation field $2z$ plus the rotation $\omega_{x_O}z$, both centred at x_O, with $\operatorname{div} v(x) = -8(b,x_O) - 8(b,z)$, $\operatorname{div}(2z) = 8$, $\operatorname{div}(\omega_{x_O}z) = 0$; the quadratic part $v(z)$

is K_b centred at x_O. The decomposition commutes with the Gaussian loop integral, the simplex integration, the special-edge factor and the expansion in ε (Thm c11-oddsimplex (ii), (iii)); the finite special vertices and the counterterms are built linearly from $S^{(1)}(y)$ and decompose in the same way. (ii) The arity-two component of $s(\hbar F_1) = \hbar\mathrm{Ob}$ at (y, P_b) has the terms $\pm\hbar F_1^{(1)}([y, P_b])$ from d_{CE}, $\pm\{S^{(1)}(y), \hbar F_1^{(1)}(P_b)\} \pm \{S^{(1)}(P_b), \hbar F_1^{(1)}(y)\}$ from the bracket with $I_{\mathfrak{psl}(4|4)}$, $(Q + \{I, -\})\hbar F_1^{(2)}(y, P_b)$, and $\{S^{(2)}(y, P_b), \hbar F_1^{(0)}\} = 0$. The bracket $\{S^{(1)}(P_b), G\}$ of a local functional $G = \int(x^\mu g_\mu + g_0)$ is $\int \partial_b$ of the fields of the density, which integrated by parts against the explicit position is minus $\int b^\mu g_\mu$, the position part. On cohomology an s-exact one-body term $\{s\phi, O\} = -s\{\phi, O\}$ for $sO = 0$ (Lemma c10-deriv). (iii) Cor c8-homotopy (ii) at (y, P_b) with $S^{(2)}(y, P_b) = 0$: the order-$\hbar^0$ part of the operator built from $\Lambda^{(2)}(y, P_b)$ is $\pm\{S^{(2)}(y, P_b), -\} = 0$, and its order-$\hbar$ part is a graded commutator with $\Lambda^{(0)}$, which on a cocycle is $\Lambda^{(0)}$-exact. Lemma c11-canonical identifies $[M, \{S^{(1)}(P_b), -\}]$ with the position part of a map M at most polynomial in the position. The particular cases use the brackets listed in (ii), with signs fixed by the order-$\hbar^0$ identity of Cor c8-homotopy (ii). □

Corollary 3.3 (Special supercharges and special conformal fields)**.** *Under the hypotheses of Theorem 3.2, for $O = \int \omega\, X_1 \cdots X_n$ at the point x_O of its density:*

$$S_1(O) = Q_1(O)\big|_{\rho(x_O)\eta} + S_1^0(O), \qquad S_1^0(O) := \sum_{a<b} \pm\Upsilon_\eta^0(X_a, X_b) - \{\hbar F_1^{(1)}(\eta)^0, O\},$$

$$K_1(O) = v(x_O)^\mu P_1(O)\big|_{P_{e_\mu}} - (b, x_O)\, D_1^0(O) + M_1^0(O)\big|_{\omega_{x_O}} + K_1^0(O),$$

with $\hbar F_1^{(1)}(y)^0$ the part of $\hbar F_1^{(1)}(y)$ centred at x_O, the superscript 0 denoting the centred parts of Theorem 3.2 (i), provided the one-body terms decompose as in Theorem 3.2 (ii); otherwise the difference is the bracket with the right side of (3.2) *minus its first term. On the scalar bilinears $\Upsilon_\eta^0(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ vanishes on the colour singlet and $\{\int s_0, A|_{W_\mathbb{R}^\perp}\} = 0$, so the two-body part of S_1 is that of Q_1 along $\rho(x_O)\eta$ (Prop c11-SKbil (i)); $\Upsilon_{K_b}^0(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ and the rotation part vanish there (Prop c11-SKbil (ii)).*

Proof. Insert (3.1) into (2.1) for $y = \eta$, K_b, and the corresponding decompositions of the one-body terms; the components along P_{e_μ}, $\rho(e_\mu)\eta$, Δ and $M_{\omega_{x_O}}$ are those of (2.1) along these generators. □

3.2 The special conformal component at a finite endpoint

Proposition 3.4 (Independence of the endpoint along K_b from the defect flow)**.** *Along K_b, at every finite T: (i) every defect cycle has $T\partial_T = O(\varepsilon)$, so Υ_{K_b} is independent of T; (ii) the graphs of the classes (Br) and (Ac) carry at fixed ε the term $\frac{3}{8\pi^2}\hbar \log T$ times a tree with the factor $e^{-\varepsilon|p|^2} - 1$ at a generic momentum, or with its gradient $-2\varepsilon p\, e^{-\varepsilon|p|^2}$ along (η_1, η_2), and tend to 0 as $\varepsilon \to 0$ at fixed T; (iii) hence $K_1(O)$ of* (2.1) *is independent of T, and the observables keep the term $\frac{3}{8\pi^2}\hbar \log T \sum_a \partial_{A|_V = b\otimes X_a}\partial_{c=X^a}$ of their order-$\hbar^0$ part generated by the flow (Thm c8-K), whose image under Q vanishes on gauge-invariant O.*

Proof. (i) Theorem 2.2 (v). (ii) Prop c8-Tsafe: the coefficient of $\log T$ of a graph with $S^{(1)}(K_b)$ floating on its cycle is $\frac{3}{8\pi^2}\hbar$ times the tree obtained by cutting the insertion out of the cycle with the freed legs on the constant modes $A|_V = b \otimes X_a$, $c = X^a$; in (Br) and (Ac) this tree contains the special edge at a generic momentum (Rem after Thm c8-K). (iii) Theorem 2.2 (v) and Thm c8-K. □

3.3 The conjugation and $K = P^\dagger$

Lemma 3.5 (Ward identity of the two-point cochain from the defect flow)**.** *Let O_1, O_2 be gauge-invariant local functionals of degree 0 with disjoint supports, t_1, t_2 square-zero parameters of their parities, and let the two-point cochain at scale T be the coefficient of $t_2 t_1$ at the zero field of $W\big(P(\varepsilon, T), J +$*

$t_1O_1(\varepsilon)+t_2O_2(\varepsilon)\big)$ *as* $\varepsilon\to 0$ *(Def c8-2pt). Assume Hypothesis 1.5 along* Y *and local representatives* $L(O_a)$ *with* $L(O_a)^{(0)}=0$ *modulo* $\hbar^2$*. Then in arity one along* Y*, modulo* $O(\hbar^2)$*,*

$$\big\langle\Lambda^{(1)}(Y)O_1,O_2\big\rangle_T+(-1)^{|Y||O_1|}\big\langle O_1,\Lambda^{(1)}(Y)O_2\big\rangle_T=B_T(Y;O_1,O_2),\tag{3.4}$$

where $\langle\,,\rangle_T$ *is the arity-zero two-point function at scale* T *and* B_T *is* $\hbar\Delta_T$ *of the quadratic part at the zero field of the arity-one two-point cochain, the graphs of its tree part closed by one kernel* K_T*.* $B_\infty=0$*; for* Y *along which the two-point cochain converges as* $T\to\infty$ *(Prop c8-Tsafe:* Σ*,* Σ^**,* Δ*, translations, rotations)* (3.4) *holds at* $T=\infty$ *with zero right side.*

Proof. By Lemma c8-transfer and Lemma c8-flow applied to $J+t_1O_1(\varepsilon)+t_2O_2(\varepsilon)$, the t_1t_2-part of d_T of the effective functional at scale T is the flow of the t_1t_2-part of the defect vertex, which is the special-edge part of $\{t_1O_1(\varepsilon),t_2O_2(\varepsilon)\}_\varepsilon$, of order $e^{-|x_1-x_2|^2/4\varepsilon}$ at separated supports and tending to 0, together with the flows in which the vertex $t_1L(O_1)$, resp. $t_2L(O_2)$, sits at the place of t_1O_1, resp. t_2O_2. At the zero field: Q is induced by a linear operator, so $(QF)|_{e=0}=0$; d_{CE} vanishes on arity zero; $\{F,G\}_T|_{e=0}$ needs one-leg parts of both arguments, which vanish for colour singlets and for the effective interaction (a one-leg part is an invariant vector of the adjoint representation); $\hbar\Delta_T$ of the t_1t_2-part leaves B_T. The right side at arity one is $\langle L(O_1)^{(1)}(Y),O_2\rangle_T\pm\langle O_1,L(O_2)^{(1)}(Y)\rangle_T$ plus the arity-one parts of the two-point cochains of $L(O_a)^{(0)}=0$. $\Delta_\infty=0$ (Def c3-delta). □

Theorem 3.6 ($K=P^\dagger$ as a consequence)**.** *Assume: (H1) a one-loop quantisation through arity one along* P_n*,* K_n *and the odd generators considered, and through arity two along* (P_n,K_n)*, reflection compatible (Thm c8-fields (iii): the average of a solution and its reflection is a solution); (H2) local representatives on a family* $\mathfrak{F}$ *of gauge-invariant cocycles at the origin of one length, stable under* $\Lambda^{(1)}(P_n)$*,* $\Lambda^{(1)}(K_n)$ *and the classical generators, with* $L_1^{(0)}=0$ *on* $\mathfrak{F}$ *for a reflection-compatible finite part* ψ*; (H3)* $\lim_{T\to\infty}B_T(K_n;O_1,O_2)=0$ *for* O_1,O_2 *in the descendants of* $\mathfrak{F}$ *entering the radial inner product (Lemma 3.5). Then on* $\mathfrak{F}$*, with the radial inner product of Def c8-inner and* $\mathrm{Gram}_{\mathrm{tree}}$*,* Gram_1 *its tree and one-loop parts,*

$$K_1\ \textit{along}\ K_{\bar b}=-\big(P_1\ \textit{along}\ P_b\big)^{\mathrm{adj}}+\hbar\big[\{S^{(1)}(K_{\bar b}),-\},\,\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big],\tag{3.5}$$

and the same form holds for S_1 *against* Q_1 *and for* D_1 *(Cor c8-gram). In a scheme with* $P_1=0$ *on* $\mathfrak{F}$ *(Def c8-radial),* $K_1=\hbar[\{S^{(1)}(K_{\bar b}),-\},\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1]$*.*

Proof. Lemma 3.5 gives the Ward identity assumed in Thm c8-adjoint for every Y along which the two-point cochain converges, and by (H3) along K_n in the limit $T\to\infty$: the left side of (3.4) involves only arity-zero two-point functions of the local functionals $\Lambda^{(1)}(K_n)O_a$, which converge as $T\to\infty$ (Prop c8-2pt (ii)), and K_1 is independent of T (Proposition 3.4). Reflection compatibility (H1), (H2) and Thm c8-fields give $(\{S^{(1)}(X),a\})^{\Theta_n}=(-1)^{1+|a||X|}\{S^{(1)}(X^{\dagger_n}),a^{\Theta_n}\}$ and its extension to $\Lambda^{(1)}$ at order $\hbar$; Prop c8-C and Def c8-radial (which uses $\Lambda^{(1)}(P_n)$, $\Lambda^{(1)}(K_n)$ and Cor c8-homotopy at (P_n,K_n), hence the arity-two part of (H1)) carry the reflection identity to the radial one. Thm c8-adjoint and Cor c8-gram then give (3.5) with $P_b^\dagger=-K_{\bar b}$ (Thm c8-dag (i)). □

Remark 3.7 (The boundary term B_T)**.** In $B_T(K_n)$ the kernel $K_T(y,y')=(4\pi T)^{-2}e^{-|y-y'|^2/4T}$ closes two legs of a tree with the vertices O_1, O_2, $S^{(1)}(K_n)$ and possibly vertices of I. At the zero field a field leg and an antifield leg must remain, and in the simplest graph the antifield leg A^* of $S^{(1)}(K_n)$ at y, joined by $P\star v.$ to O_1, is contracted with a letter of O_2 at x_2: the integrand is of order $|y|^{-1}$ (the factor $|y|^2$ of v against one derivative of an edge of order $|y|^{-2}$) and $\int_{|y|^2\lesssim T}|y|^{-1}T^{-2}d^4y=O(T^{-1/2})$. Hypothesis (H3) is a statement about finitely many tree graphs on each family; it is checked by the engines (gate G-KPbdry) and is not used for any number. Theorem 3.6 is a cross-check of K_1 computed by (2.1) along K_b; it never defines K_1.

4 The two-body locality of the composite term

Throughout this section $x \in \{\Delta, P_b, M_\omega\}$, so that the composite line L_x is Q-closed (Lemma 1.4 (ii)), $T = \infty$ with generic momenta or any finite T, and $P = P(\varepsilon, T)$. The composite term enters only the second evaluation (Theorem 2.8) and the identities of Theorem 2.9; the components themselves are given by Theorem 2.2, which needs no composite term and holds along every generator.

Definition 4.1 (Composite loops of the action)**.** $\Xi_x(\varepsilon)$ is the sum of the amputated one-particle-irreducible composite loops of I along x: cycles of vertices of I and edges P in which one edge is the composite line L_x, with all other legs open; equivalently $\frac{\hbar}{2}\partial_{\mathsf{L}_x} W(P, I)_{\mathrm{tree}}$ is the tree flow of $\Xi_x(\varepsilon)$ with edges P and vertices I. $\Xi_x^\sigma(\varepsilon)$ is the sum of the same loops in which one edge of the cycle other than L_x is the special edge, with the sign of Prop c11-master (minus the loops with one edge $K_\varepsilon - K_0$). The renormalised composite functional is $\Xi_x^{\mathrm{ren}} := \lim_{\varepsilon\to 0}\big(\Xi_x(\varepsilon) - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)\big)$, a nonlocal functional of degree -1 at generic momenta; its quadratic part along Δ is the renormalised composite two-point functional of Rem c9-Pi. Composite graphs of a local functional O (Def 2.7) are of type I when the cycle passes through two letters of O and of type II when it passes through vertices of I alone; type II is the tree insertion of $\Xi_x(\varepsilon)$ into O, written $\mathrm{tr}(\Xi_x, O)$.

Lemma 4.2 (Images under Q of the composite loops)**.** *For every $\varepsilon > 0$, as functionals at generic momenta,*

$$s\,\Xi_x(\varepsilon) = \Xi_x^\sigma(\varepsilon).$$

Consequently: (i) Ξ_x^σ has no term $\log\varepsilon$ and no poles, $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) = \log\varepsilon\cdot L^{(1)}(x)$ with $s\,L^{(1)}(x) = 0$; (ii) the composite part along x of Lemma 2.6 is $s\,\Xi_x^{\mathrm{ren}} = [\varepsilon^0]\Xi_x^\sigma$, a local functional, although Ξ_x^{ren} is nonlocal on the letters $A|_V$, λ, c.

Proof. Compute Q of the tree flow of Ξ_x in two ways. First, $\frac{\hbar}{2}\partial_{\mathsf{L}_x} W(P, I)_{\mathrm{tree}}$ is the tree flow of Ξ_x, and ∂_{L_x} anticommutes with Q (Lemma 1.4 (ii), Lemma c9-ops), so by Prop c4-defectflow at order $\hbar^0$ and the classical master equation, Q of it is $-\frac{\hbar}{2}\partial_{\mathsf{L}_x}$ of the tree flow of $\frac{1}{2}\{I, I\}_\varepsilon^\sigma$: the composite closures of the trees of I with one special edge (Prop c11-master with $M = 0$). Sort them by the position of the special edge: on the cycle closed by L_x, which gives the tree flow of Ξ_x^σ, or on a tree, which gives the tree flow of Ξ_x with one special edge on a tree edge, between two vertices of I or between Ξ_x and a vertex of I. Second, with Ξ_x as a vertex, Prop c4-defectflow applied to $I + \delta\,\Xi_x$, δ of degree $+1$, gives the tree flow of $s\Xi_x$ plus the tree flows with the special-edge parts of $\{I, \Xi_x\}_\varepsilon$ and of $\frac{1}{2}\{I, I\}_\varepsilon$ on tree edges. The graphs with the special edge on a tree are the same in both evaluations with the same signs, the sign rule of Prop c11-master being the Koszul rule of the second evaluation. Hence the tree flows of $s\Xi_x$ and of Ξ_x^σ agree, and the tree flow is injective on functionals graded by the number of legs (the argument of Thm c8-L (i), which uses only the leg count). (i) Ξ_x^σ is a sum of cycles with one special edge and the composite line, which by Thm c11-simplex (whose proof applies to cycles of vertices of I) are $\sum_m \varepsilon^{-m} a_m + a_0 + o(1)$ without $\log\varepsilon$. The poles of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ vanish by the weight count of Theorem 2.2 (iv), so $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) = \log\varepsilon\, L^{(1)}(x)$, and the coefficient of $\log\varepsilon$ in $s\Xi_x = s\,I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) + s\Xi_x^{\mathrm{ren}} + o(1)$ gives $sL^{(1)}(x) = 0$, while the poles of Ξ_x^σ vanish. (ii) The coefficient of ε^0. □

Lemma 4.3 (Special edges on trees)**.** *Let O be a local functional. (B) The composite graphs of type I with the special edge on a tree (at an output of the cycle or at a spectator letter of O), together with the special-edge parts of the counterterm trees of $\mathrm{ct}_x^{(1)}(O)$, are $e^{-\varepsilon|p|^2} - 1$ at a generic momentum p times renormalised composite cycles, and tend to 0. (D) The composite graphs of type II with the special edge on a tree tend to 0, and so do the special-edge parts of the counterterm trees of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$.*

Proof. (B) The special edge on a tree edge of momentum p factorises: the graph is $(e^{-\varepsilon|p|^2}-1)$ times the unrenormalised composite cycle through the pair with trees; the counterterm vertex $-\mathrm{ct}_x^{(1)}(O)[\varepsilon]$ joined by the same special edge gives $(e^{-\varepsilon|p|^2}-1)$ times minus its singular part, so the sum is $(e^{-\varepsilon|p|^2}-1)$ times the renormalised composite cycle, which has a limit at generic momenta (Prop c9-structure (i) and its extension by Lemma 1.4). At a spectator letter of O the same holds with the special-edge part of $\{I,O\}_\varepsilon$ and of $\{I,\mathrm{ct}_x^{(1)}(O)\}_\varepsilon$. (D) The graphs are $(e^{-\varepsilon|p|^2}-1)$ times $\Xi_x(\varepsilon)$, which is $\log\varepsilon\, L^{(1)}(x)$ plus a convergent functional (Lemma 4.2 (i)); $(e^{-\varepsilon|p|^2}-1)\log\varepsilon\to 0$. The counterterm trees carry $(e^{-\varepsilon|p|^2}-1)\log\varepsilon$. □

Proposition 4.4 (The image of the cycles through the pairs)**.** *For every local functional O and every $\varepsilon>0$,*

$$Q(\textit{type I composite graphs of } O) = \text{(A)} + \text{(B)} - \epsilon_\Xi\{\Xi_x(\varepsilon), O\} - (\textit{composite graphs of } sO),$$

where (A) are the special-edge composite cycles of the pairs of letters of O with trees of I on their outputs (Def c11-sp), (B) the type I graphs with the special edge on a tree, $\epsilon_\Xi\in\{\pm1\}$ the sign with which $\{\Xi_x,O\}$ enters $Q(\mathrm{tr}(\Xi_x,O))$, and the last term is present only for $sO\neq 0$:

$$Q\,\mathrm{tr}(\Xi_x,O) = \mathrm{tr}(s\Xi_x,O) + \epsilon_\Xi\{\Xi_x,O\} \pm \mathrm{tr}(\Xi_x,sO) + \text{(D)}.$$

Proof. By Prop c11-master the image under Q of all composite graphs of O is $-\frac{\hbar}{2}\partial_{\mathsf{L}_x}$ of the trees with one vertex sO or one special edge; the trees with one vertex sO closed by L_x are the composite graphs of sO, of both types. The special edge lies on the cycle through two letters (A), on a tree of a type I graph (B), on the cycle of a type II loop (C), or on a tree of a type II graph (D); by definition (C) is $\mathrm{tr}(\Xi_x^\sigma,O)$. The image of type II is computed with Ξ_x as a vertex by Prop c4-defectflow applied to $I+\delta_1\Xi_x+\delta_2 O$, $\delta_1\delta_2$-part: the vertex $s\Xi_x$ with O, the vertex sO with Ξ_x, the direct contraction $\{\Xi_x,O\}$, and the special edges on trees, which are (D); there is no other term. Subtract: by Lemma 4.2, $\mathrm{tr}(s\Xi_x,O)=\mathrm{tr}(\Xi_x^\sigma,O)=$ (C), so (C) cancels, and $\mathrm{tr}(\Xi_x,sO)$ is the type II part of the composite graphs of sO. □

Theorem 4.5 (The composite term is two-body and local)**.** *Let $O=\int\omega\, X_1\cdots X_n$ be a monomial local functional, $x\in\{\Delta,P_b,M_\omega\}$.*

- *(i) The special-edge composite cycles of every pair are $\sum_{m\geq1}\varepsilon^{-m}$ times local coefficients, plus the composite pair kernel $\mathcal{C}_x(X_a,X_b)$, plus a remainder tending to 0; no $\log\varepsilon$.*
- *(ii) $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)=\log\varepsilon\, L^{(1)}(x)$ with $sL^{(1)}(x)=0$, and the composite part along x is $s\Xi_x^{\mathrm{ren}}=[\varepsilon^0]\Xi_x^\sigma$.*
- *(iii) (Arity-one BRST covariance.) If $sO=0$: $s\,\mathrm{lc}_x^{(1)}(O)=-\epsilon_\Xi\{L^{(1)}(x),O\}$, and $\sum_{a<b}\pm[\varepsilon^{-m}](\textit{special-edge composite cycles of } (X_a,X_b))=s\,\mathrm{ct}^{(1)}_{x,m}(O)$ for $m\geq1$.*
- *(iv) If $sO=0$, the composite term of O along x (Def 2.7) exists and*

$$(\textit{composite term of } O \textit{ along } x) = \sum_{a<b}\pm\mathcal{C}_x(X_a,X_b), \tag{4.1}$$

  a local two-body functional with no one-body term.
- *(v) (Cancellation of the nonlocal one-body terms.) If $sO=0$, the image under Q of the renormalised composite cycles through the pairs (type I) contains $-\epsilon_\Xi\{\Xi_x^{\mathrm{ren}},O\}$, and the image of the renormalised composite loops on the legs (type II) contains $+\epsilon_\Xi\{\Xi_x^{\mathrm{ren}},O\}$; both are nonlocal one-body terms, on the letters $A|_V$, λ, c (the factors $\log|p|^2$ and, on $A|_V$, $(p,a)(p,b)|p|^{-2}$ of Rem c9-Pi), and on $A|_{W_{\mathbb{R}}^\perp}$ local; they cancel in the composite term. In the form of Thm c10-D1, the composite terms of the pairs plus the composite terms of the legs equal $\sum_{a<b}\pm\mathcal{C}_x(X_a,X_b)$.*

- *(vi) (Gauge-variant operators.) For $sO \neq 0$ the right side of* (4.1) *acquires minus the limit of the renormalised composite graphs of sO, both types, minus their counterterm trees: the nonlocal brackets $\{\Xi_x^{\mathrm{ren}}, -\}$ on the letters of sO and the Q-preimages of the composite cycles through the pairs of letters of sO. The components of O are nevertheless local and two-body by Theorem 2.2.*

Proof. (i) Thm c11-simplex applied to each chain of Def c11-sp. (ii) Lemma 4.2.

(iv) and (iii). The composite term is $\lim_{\varepsilon\to 0}$ of Q of the composite graphs minus their counterterm trees, minus the tree insertion of the composite part. Write the counterterm trees as $-\mathrm{tr}(\mathrm{ct}_x^{(1)}(O)[\varepsilon])$ and $-\mathrm{tr}(I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O)$. By Prop c4-defectflow and Lemma c6-bvexact, with $sO = 0$,

$$\begin{aligned} Q\big(-\mathrm{tr}(\mathrm{ct}_x^{(1)}(O)[\varepsilon])\big) &= -\mathrm{tr}\big(s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]\big) + (\text{special edges}),\\ Q\big(-\mathrm{tr}(I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O)\big) &= -\mathrm{tr}\big(s\, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O\big) - \epsilon_\Xi\{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O\} + (\text{special edges}), \end{aligned}$$

the sign of the bracket being the same as for Ξ_x, since both enter as vertices of degree -1 joined to O by a tree. By Proposition 4.4 for type I and its displayed formula for type II, the sum is

$$\text{(A)}+\text{(B)}+\text{(D)}+(\text{special edges})+\mathrm{tr}\big(s(\Xi_x - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)), O\big) - \epsilon_\Xi\{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O\} - \mathrm{tr}\big(s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]\big),$$

the brackets $\pm\epsilon_\Xi\{\Xi_x, O\}$ of the two types cancelling. By Lemma 4.3, (B), (D) and the special edges tend to 0. By Lemma 4.2, $\mathrm{tr}(s(\Xi_x - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)), O) \to \mathrm{tr}(s\Xi_x^{\mathrm{ren}}, O)$, the tree insertion of the composite part, which is subtracted in the definition. What remains is (A) $-\epsilon_\Xi \log\varepsilon\{L^{(1)}(x), O\} - \mathrm{tr}(s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon])$. The left side has a limit (the renormalised composite graphs converge and Q multiplies by polynomials in the momenta), (A) has no $\log\varepsilon$ by (i): the coefficient of $\log\varepsilon$ gives $\mathrm{tr}(s\mathrm{lc}_x^{(1)}(O) + \epsilon_\Xi\{L^{(1)}(x), O\}) = 0$, and the coefficient of ε^{-m} gives $\mathrm{tr}\big(\sum_{a<b} \pm a_m(X_a, X_b) - s\,\mathrm{ct}^{(1)}_{x,m}(O)\big) = 0$; the trees with these local vertices are the same at every ε, so they cancel identically, and injectivity of the tree flow gives (iii). At order ε^0 the trees on the outputs of (A) converge to trees with edges $P(0,T)$, and the limit is the tree flow of $\sum_{a<b} \pm\mathcal{C}_x(X_a, X_b)$; Thm c8-L (i) gives (4.1).

(v) Proposition 4.4 and its displayed formula, with the renormalisation of the previous step: type I contains $-\epsilon_\Xi\{\Xi_x, O\}$, type II $+\epsilon_\Xi\{\Xi_x, O\}$, and the counterterm trees move Ξ_x to Ξ_x^{ren} in the second and add $-\epsilon_\Xi \log\varepsilon\{L^{(1)}(x), O\}$, which (iii) removes from the first. The renormalised composite loops on a leg X give the bracket of Ξ_x^{ren} with X; its quadratic part along Δ is the functional of Rem c9-Pi, local on $A|_{W_{\mathbb{R}}^\perp}$ and containing $\log|p|^2$ on $A|_V$, λ, c. The sum of the images of the two types has no bracket, which is (iv). The form of Thm c10-D1 splits the same sum into the images of the renormalised composite cycles through the pairs (the composite terms of the pairs) and the brackets $\epsilon_\Xi\{\Xi_x^{\mathrm{ren}}, X_a\}$ plus the tree insertion of $s\Xi_x^{\mathrm{ren}}$ (the composite terms of the legs and the composite part).

(vi) For $sO \neq 0$, Prop c11-master keeps the trees with one vertex sO, whose closures by L_x are the composite graphs of sO (Proposition 4.4); the displayed formula for type II keeps $\pm\mathrm{tr}(\Xi_x, sO)$, the type II part of these. With their counterterm trees they converge to the renormalised composite graphs of sO, nonlocal through $\{\Xi_x^{\mathrm{ren}}, -\}$ on the letters of sO. □

Remark 4.6 (Rotations, and the generators without a closed line)**.** For M_ω the composite line is the term $\int(\omega x\cdot\partial A|_V + \omega A|_V, A^*|_V)$ joined to two edges, Q-closed because $M_\omega.$ commutes with Q^{GF} and H; every step above applies, and by Theorem 3.2 (i) $\mathcal{C}_{M_\omega} = (\omega x_O)^\mu \mathcal{C}_{P_{e_\mu}} + \mathcal{C}^0_{M_\omega}$, the intrinsic part computed with the spin term $\frac{1}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}$ and the transport $\omega z\cdot\partial$ centred at x_O. Along K_b, ξ and η the composite line is not Q-closed (Lemma 1.4), Prop c11-master fails for it, and there is no composite term in the sense of Def 2.7; the components along these generators are local and two-body by Theorem 2.2, which is the strict statement in all cases.

Remark 4.7 (What the locality uses)**.** The proof uses $sO = 0$ exactly (or keeps sO, (vi)), the classical master equation, $[x., Q] = 0$ with $[Q, P\star x.P\star] = 0$, the injectivity of the tree flow on functionals graded

by legs, the absence of $\log \varepsilon$ on cycles with one special edge, and the weight count that excludes poles of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$. It replaces the outline of Thm c11-local: step (C) is Lemma 4.2 with Proposition 4.4, step (D) is Lemma 4.3, and the cancellation of the nonlocal one-body terms is the cancellation of the brackets $\pm\epsilon_\Xi\{\Xi_x, O\}$ between the two types, forced by Prop c11-master, whose right side contains no bracket. The sign ϵ_Ξ never needs to be evaluated.